



Huawei Mate 30 Pro: 4 cameras?

Leaked images suggest that the Huawei Mate 30 Pro will have four cameras.
<https://digit.in/sep19-67>



100 million subscribers

PewDiePie becomes first solo YouTube creator to hit 100 million subscribers.
<https://digit.in/sep19-69>

ZOTAC RTX 2070 Super AMP Extreme

More competition in the mid-range

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he RTX Super cards were announced to counter AMD's RX 5700 series of graphics cards. Not that the NVIDIA cards managed to provide more performance than the AMD cards, it did however, bring the prices down in the mid-range segment. After the ridiculous pricing that was introduced with the RTX series, this correction was a much needed one.

The Super cards have a few extra Streaming Multiprocessors (SMs) over the non-super cards. With the 2070 and the 2060 Super cards, we have access to 4 extra SMs whereas with the 2080 Super, we only get 2 extra SMs. Needless to say, the impact is much more pronounced with the 2060 Super and 2070 Super than with the 2080 Super cards. The ZOTAC RTX 2070 Super AMP Extreme is a custom card with additional cooling and a slight bump in the boost frequency. The boost with partner cards starts at 1770 MHz and at 1830 MHz, this particular card has a 60 MHz bump. The highest we've seen is the ASUS STRIX GAMING OC with 1935 MHz, so the best binned GPUs have a bit more headroom.

Build Quality

The card has a triple fan setup and the shroud isn't as extravagant as we've seen in some of the older ZOTAC cards. There is no visible lighting on the front side. At the top, we see the only lighting element in the form of the "ZOTAC Gaming" label. Under the shroud, we see a nice and thick fin stack with five nickel-plated heat pipes. The contact surface of the heat-sink with the GPU is all copper. Then we have two thick metal strips that run along the length of the finstack and these provide some much-needed rigidity. Underneath the heatsink stack, there's yet another black metal bracket for additional reinforcement as well



Price
₹47,290



PERFORMANCE.....81
BUILD79
VALUE FOR MONEY....75

as for dissipating the heat from the MOSFETs.

As for the power connectors, it uses one 6-pin and one 8-pin connector which means it can pull up to 300 Watts. There are no LED indicators near the power connectors.

The card uses a uP9512R controller for controlling the VRM circuitry for the GPU VRM as well as for the memory VRM. The uP9512R can be configured up to 8-phases and supports NVIDIA Open Voltage Regulator type 4i+ with PWMVID. The memory VRM uses another uP9512R, as stated earlier, but it's configured for 2-phases.

Speaking of memory, the ZOTAC RTX 2070 Super AMP Extreme uses Micron chips for the VRAM. The ZOTAC RTX 2080 Super, on the other hand, has Samsung chips for the memory. Using Micron, in a way, limits the OC capabilities of the device. Lastly, there is no "Power Boost" capacitors on this PCB which is usually found on the higher-end cards. They help filter the power delivered to the GPU.

Performance

We started off by running a couple of games at different resolutions. Far Cry 5 ran at 135 FPS on FHD, 108 on 1440p and a little over 60 FPS on 4K. In Shadow of the Tomb Raider, we managed to get about 121 FPS on FHD, 83 FPS on 1440p and 45 FPS on 4K. Compared to the RX 5700 XT, the

RTX 2070 Super is better by a narrow margin and your choice of graphics card simply comes down to whichever has the better price tag. Both companies have ramped up their

efforts to push out frequent driver updates to coincide with game launches so there really isn't much of a differentiating factor aside from RTX support on the NVIDIA cards.

Verdict

The ZOTAC RTX 2070 Super AMP Extreme is certainly a powerful graphics card which competes in the mid-range segment in terms of performance. However, with AMD's recent launches, and the price-to-performance factor is where the RTX 2070 Super, in general, takes a beating. The additional 13K is a bit too much to ask for the meagre performance difference that the 2070 Super brings to the table. ZOTAC's 5-year warranty has always been a key factor in buying decisions and that doesn't change with this card either. If you want a great mid-range card with support for RTX, then this is a great candidate.

—Mithun Mohandas

SPECIFICATIONS

CUDA CORES: 2560 | ROPS: 64 | CORE CLOCK: 1605 MHz | BOOST CLOCK: 1830 MHz | MEMORY CLOCK: 1750 MHz | MEMORY BUS WIDTH: 256-bit | VRAM: 8 GB | TDP: 215 W | WARRANTY: 5 Years.

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